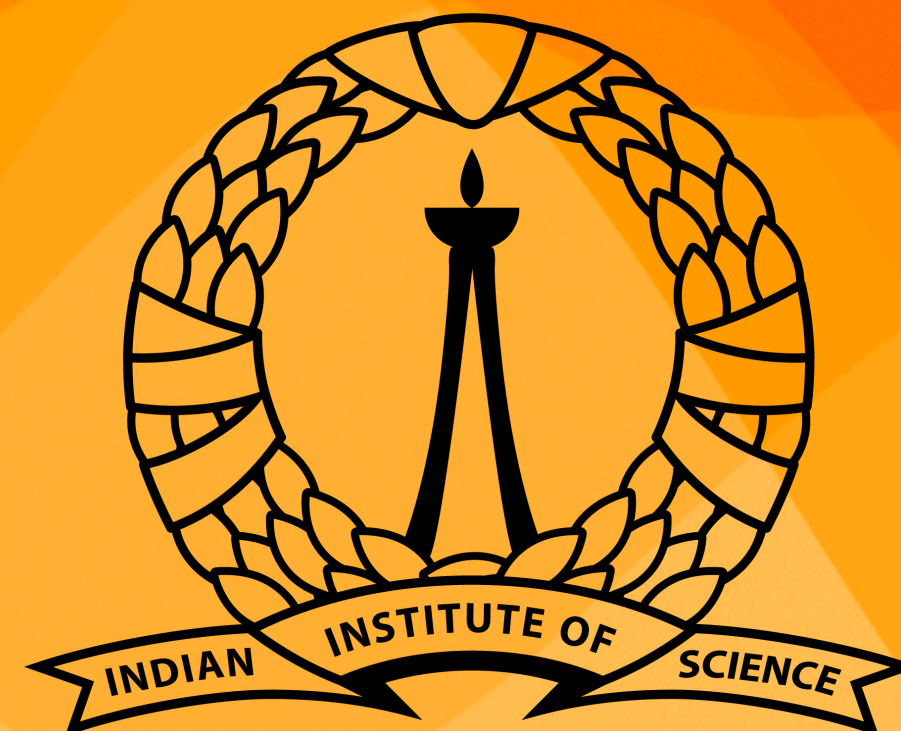


Impact of Control Access Protocols in IRS-Aided Wireless Systems

Yashvanth. L,
Dept. of ECE, IISc



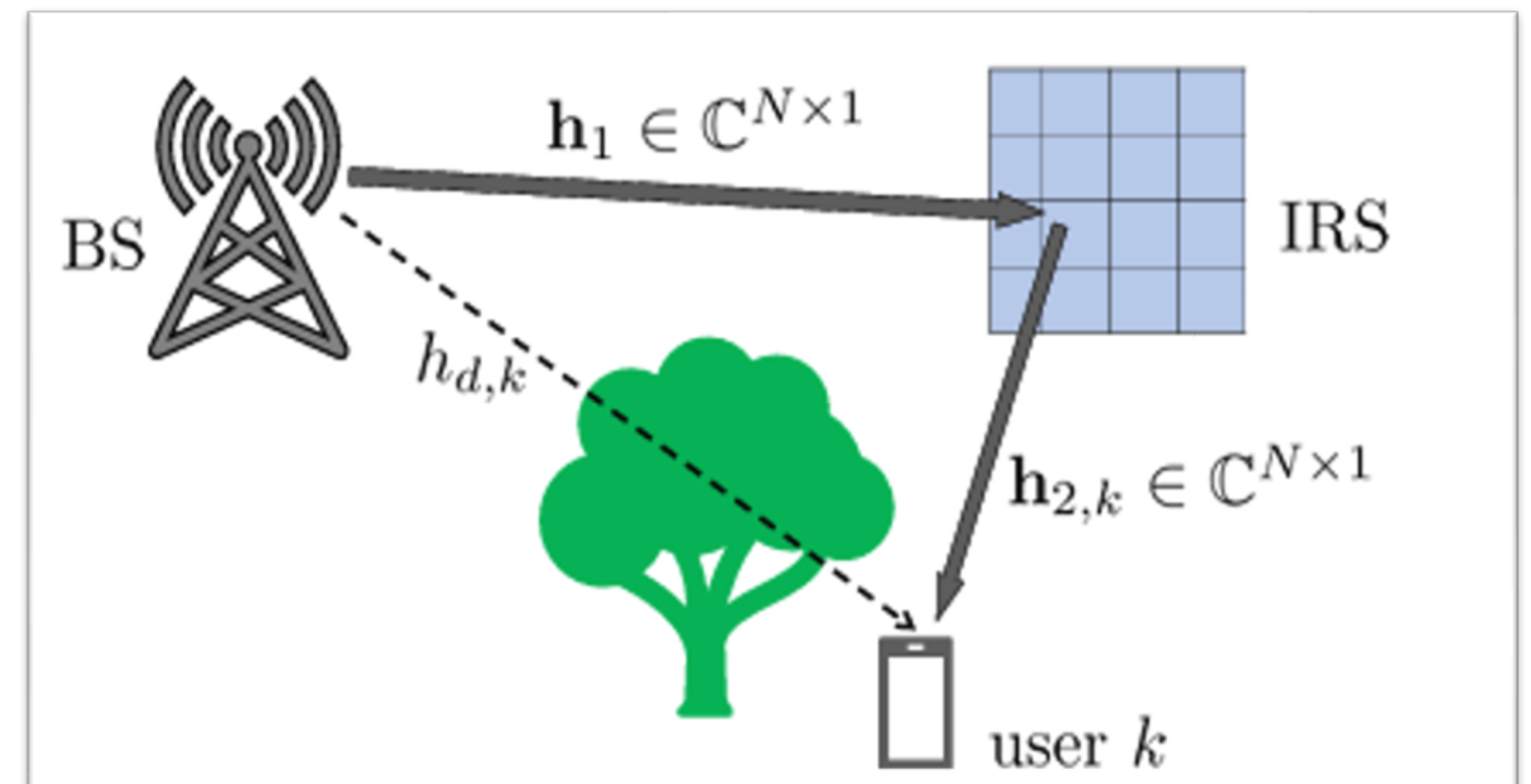
भारतीय विज्ञान संस्थान

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Basics of Intelligent Reflecting Surfaces (IRSs)

IRSs modify the wireless channel

- IRSs are capable of modifying the wireless channels
- Electronically programmable control of propagation paths
 - ➔ Adjust the reflection coefficients of the IRS elements
- Additional overheads
 - ➔ Channel estimation of all links
 - ➔ IRS phase optimization
 - ➔ Phase transportation



Perks and Open Question

- IRS/RISs Promises

- ➔ Low-power consumption
- ➔ Boosted coverage
- ➔ Better data-rates
- ➔ Improved sensing and localization

- Many works on rate analysis, beamforming, resource optimization, channel estimation, ...

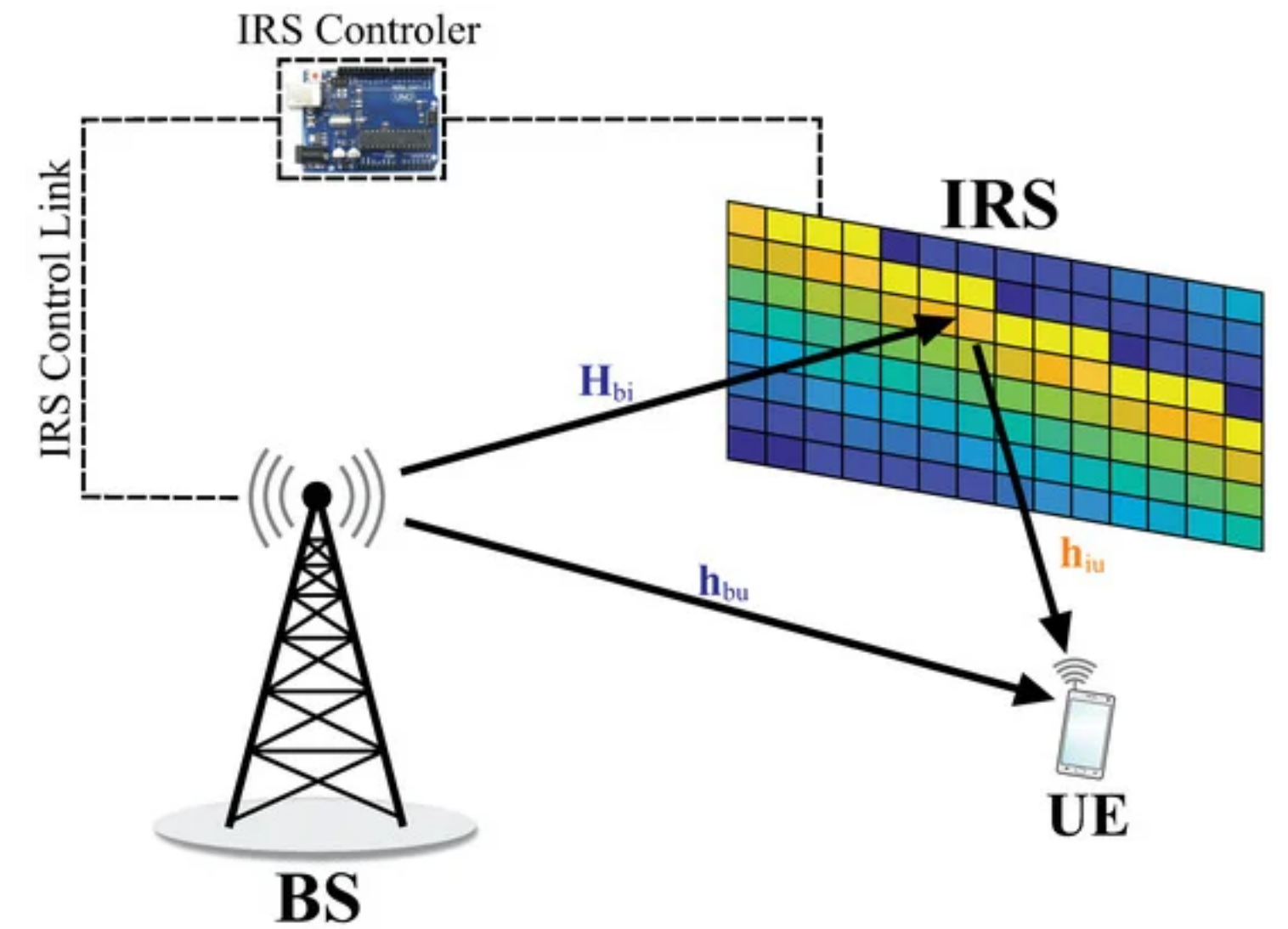
- ➔ All of them correspond to data txn phase only!

- Less discussed question:

- ➔ How do we ensure these promises, considering the impact of control overheads?
 - ✓ How do rate, reliability, and overhead change due to the CCs?
- ➔ Implementation specific presentation!

IRS Operation Model

- Single antenna BS and UE!
- IRS: IRS panel + IRS Controller (IRSC)
 - ➔ IRS panel contains N reflecting elements
 - ➔ IRS vector: $\phi = [e^{j\varphi_1}, e^{j\varphi_2}, \dots, e^{j\varphi_N}]^T \in \mathbb{C}^N$
- Switching time for a new configuration: $\tau_s \in \mathbb{R}_+$
- Common codebook at IRSC & BS: $\mathcal{C} = \{\phi_1, \phi_1, \dots, \phi_C\}, |\mathcal{C}| = C$
- IRS-control channel (IRS-CC): control link between BS and IRS
 - ➔ The control can include an index of a code OR a new IRS configuration



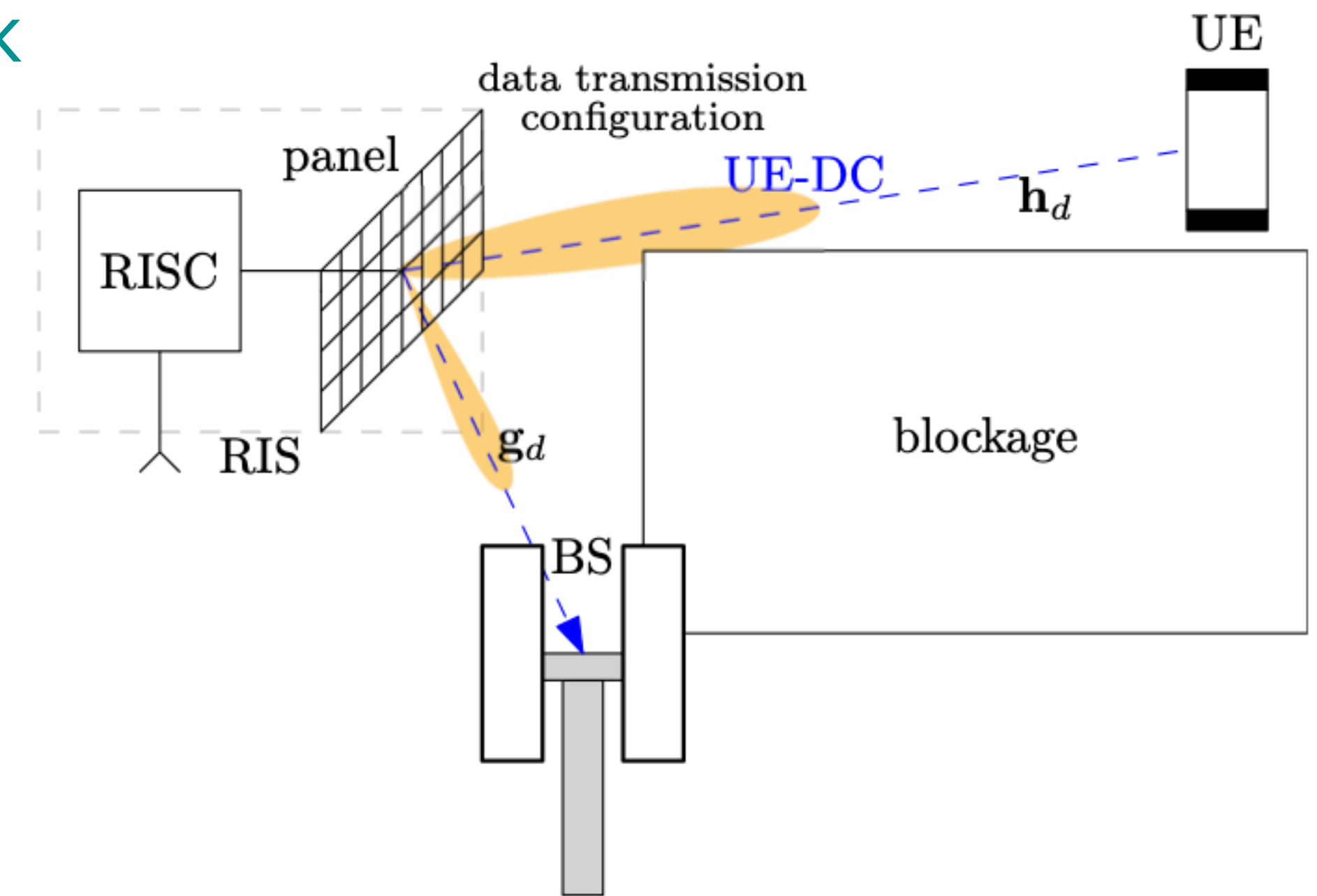
UE Communication Model

- Frame-based comm., duration in secs. : $\tau \in \mathbb{R}_+$
- Each Frame: **Control** over UE-CC & **data** over on UE-DC
- UE talks to BS (and vice-versa) on DC and CC **only through the IRS!**
- Perfect time/frequency **synchronization** across devices
- IRS ctrl config: The configuration that IRS uses in **idle mode**
 - ➡ Designed to serve **UE-CC very reliably**.
 - ➡ E.g., wide-beams that offer **reliability** with **low data rates**

Channel Models

- **Block fading** channels: longer coherence time **increases frame** duration
- Three narrowband channels in the system
 - ➔ UE-CC: BS gives control messages to **UE in downlink**
 - ➔ UE-DC: UE transmits **data in uplink**
 - ➔ IRS-CC: **BS instructs IRS** for programming
- UE-DC channels - frequency f_d & bandwidth B_d
 - ➔ UL SNR:

$$\gamma = \frac{\rho_u}{\sigma_b^2} |\phi^T (\mathbf{h}_d \odot \mathbf{g}_d)|^2 = \frac{\rho_u}{\sigma_b^2} |\phi^T \mathbf{z}_d|^2$$



CC Channel Model

◎ UE-DC channels - frequency f_u & bandwidth B_u

➔ UE-CC channel: $h_{cu} = \phi_{ctrl}^T(\mathbf{h}_c \odot \mathbf{g}_c) = \phi_{ctrl}^T \mathbf{z}_c$

➔ No LOS: $h_{cu} \sim \mathcal{CN}(0, \tilde{\lambda}_u)$, instant. SNR: $\Gamma_u = \frac{\rho_b}{\sigma_u^2} |h_{cu}|^2 \sim \text{Exp} \left(\frac{\sigma_u^2}{\rho_b \tilde{\lambda}_u} \right)$

➔ UE-CC & UE-DC use same frequency $f_d = f_u$, as IRS config. is common tuned to a freq.

◎ IRS-CC channels - frequency f_r & bandwidth B_r

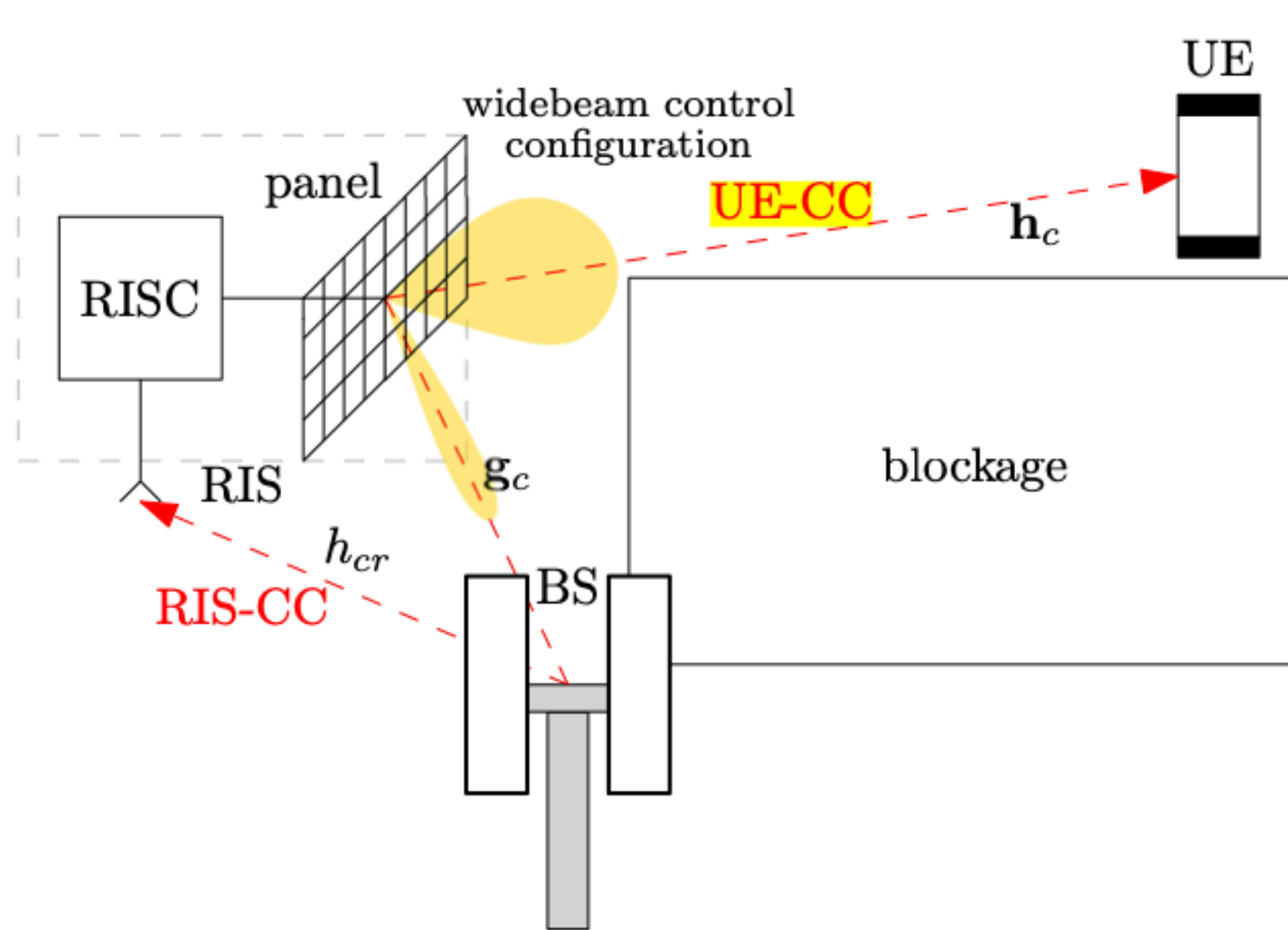
◆ Ch.: $h_{cr} \sim \mathcal{CN}(0, \tilde{\lambda}_r)$, instant. SNR @ IRS-CC RF chain: $\Gamma_R = \frac{\rho_b}{\sigma_r^2} |h_{cr}|^2 \sim \text{Exp} \left(\frac{\sigma_r^2}{\rho_b \tilde{\lambda}_r} \right)$

◆ IRS-CC can be in-band / out-of-band!

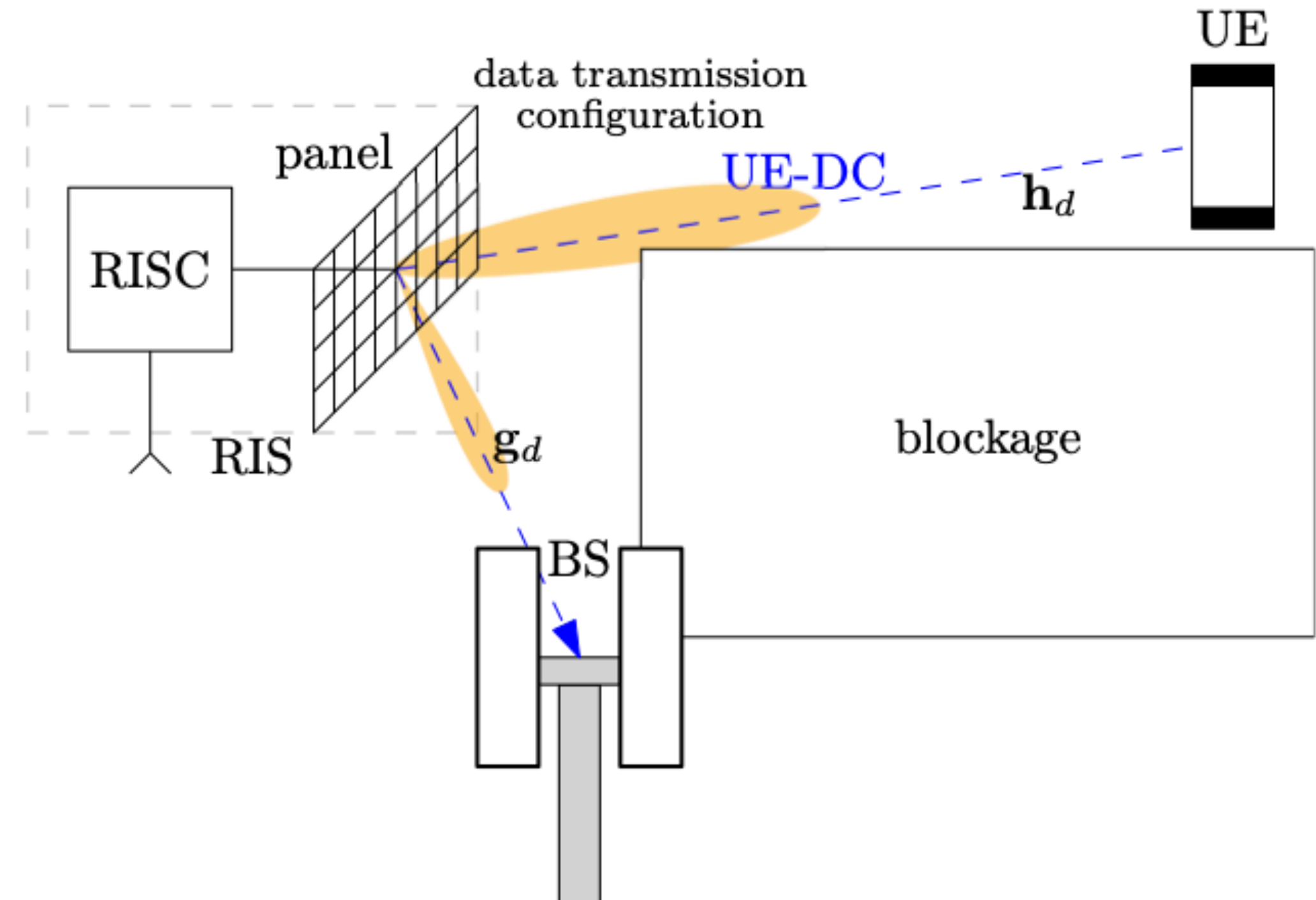
➔ In-band (IB): IRS-CC shares frequency with UE-CC: $f_d = f_R$

➔ Out-of-band (OB): Can be a separate wired link: $\lambda_r \rightarrow \infty$

CC and DC channels summary



(a) Control flow.



(b) Data flow.

The BS informs:

1. The UE to prepare payload and send over UE-CC
2. IRS to load config. over IRS-CC

The UE transmits:

The payload towards the BS through the RIS over UE-DC

IRS Enabled Communication Paradigms

The two paradigms & Signalling phases

- ◎ **Two** paradigms:

- ➔ **Optimization** based on channel estimation (OPT-CE) - enables **multiplexing**
- ➔ Codebook-based **beam sweeping** (CB-BSW) - enables **diversity** for a target KPI set by BS

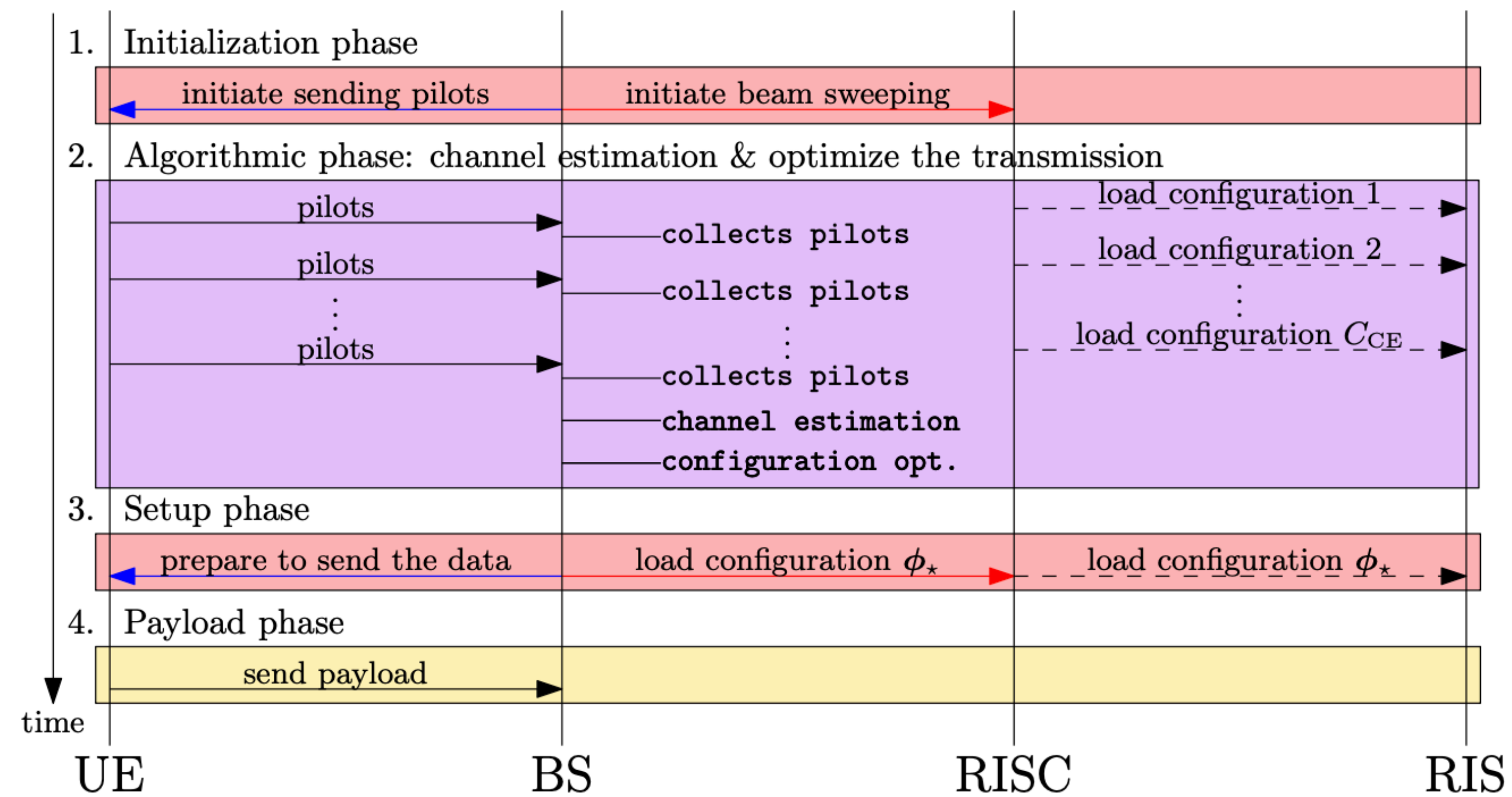
- ◎ **Four** signaling phases:

- ➔ **Initialization**: BS notifies IRS/UE about beginning of frame over CCs. Also, a RA, if needed
- ➔ **Algorithmic**: Optimization of system parameters: e.g., pilot-based ch. estim, determining IRS config. for data txn. Prone to “**algorithmic errors**”!
- ➔ **Setup**: BS informs the IRS about the config. and final confirmation with UE, e.g., the MCS
- ➔ **Payload**: Actual data txn from UE to BS

- ◎ Total frame duration: $\tau = \tau_{ini} + \tau_{alg} + \tau_{set} + \tau_{pay}$

Optimization based on Ch. estim. (OPT-CE)

- BS optimally chooses IRS config., and the CSI information is collected using pilots
- Signaling phases:



(a) OPT-CE paradigm.

Performance of OPT-CE

● **Pilot** used by UE: $\boldsymbol{\psi} \in \mathbb{C}^p$, $\|\boldsymbol{\psi}\|^2 = p$ ($p \geq 1$ for multiple UEs (or) $\tau_s \geq T_s$ (or) denoising)

● **Received signal** at BS: $\mathbf{y}_c^T = \sqrt{\rho_u} \boldsymbol{\phi}_c^T \mathbf{z}_d \boldsymbol{\psi}^T + \tilde{\mathbf{w}}_c^T \in \mathbb{C}^{1 \times p}$

● **De-correlation** at BS:

$$y_c = \frac{1}{\sqrt{\rho_u p}} \mathbf{y}_c^T \boldsymbol{\psi}^* = \boldsymbol{\phi}_c^T \mathbf{z}_d + w_c \in \mathbb{C}$$

● **Collecting the received symbols** across C_{CE} pilot transmissions: $\mathbf{y} = \boldsymbol{\Theta}^T \mathbf{z}_d + \mathbf{w}$

● **Least-squares** estimation of cascaded channels

➔ Requirement: $C_{CE} \geq N$.

➔ Choice of $\boldsymbol{\Theta}$ should be **full-column rank**. E.g., columns of DFT matrix

● Figuring out the **optimal IRS config**. $\boldsymbol{\phi}_* = \max_{\boldsymbol{\phi}} \{|\boldsymbol{\phi}^T \hat{\mathbf{z}}_d|^2 \mid |\phi_n| = 1\}$

● **Achievable SE**: $\eta_{CE} = \log_2(1 + \hat{\gamma}_{CE}) = \log_2 \left(1 + \frac{\rho_u}{\sigma_b^2} |\boldsymbol{\phi}_*^T \hat{\mathbf{z}}_d|^2 \right)$

Algorithmic Errors in OPT-CE

- If the SE selected is greater than the actual channel capacity - it leads to an outage
- Outage prob. is $p_{ae} = \mathcal{P} \{ \eta_{CE} = \log_2(1 + \hat{\gamma}_{CE}) \geq \log_2(1 + \gamma_{CE}) \}$
- This translates to

$$p_{ae} = \mathcal{P}(\hat{\gamma}_{CE} \geq \gamma_{CE}) = \mathcal{P}(|\phi_*^T \mathbf{z}_d + \phi_*^T \mathbf{n}|^2 \geq |\phi_*^T \mathbf{z}_d|^2)$$

- The exact expression depends on the ch. model!

Ch. estim. Error



Codebook-based Beam Sweeping (CB-BSW)

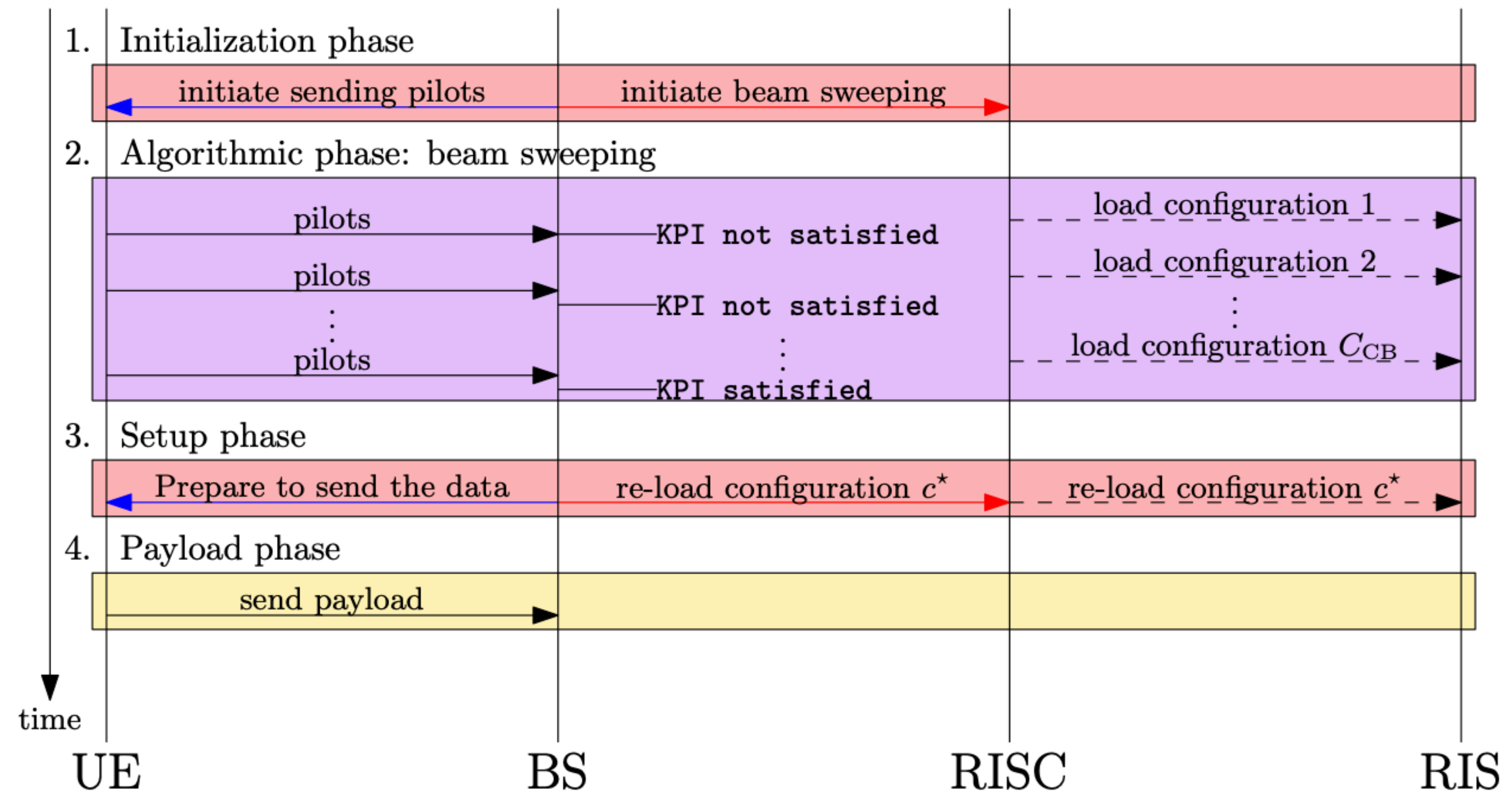
● BS **does not** require any **explicit CSI** of the UE.

● Frame types:

➔ **Fixed**: BSW process runs through **complete CB**

➔ **Flexible**: BSW process ends when **KPI is met** @ BS

➔ BS **informs termination** to IRS & UE on IRS-CC & UE-CC



(b) CB-BSW paradigm.

Performance of CB-BSW

- The uplink KPI can be a **target SNR**, say γ_0
- The goal is to find an **IRS config. that supports this SNR**
- **Received** signal at BS : $\mathbf{y}_c^T = \sqrt{\rho_u} \boldsymbol{\phi}_c^T \mathbf{z}_d \boldsymbol{\psi}^T + \tilde{\mathbf{w}}^T \in \mathbb{C}^{1 \times p}$
- **De-correlating** with pilots: $y_c = \frac{1}{p} \mathbf{y}_c^T \boldsymbol{\psi}^* = \sqrt{\rho_u} \boldsymbol{\phi}_c^T \mathbf{z}_d + w_c \in \mathbb{C}$
- **SNR** of this scheme: $\hat{\gamma}_c = \frac{|y_c|^2}{\sigma_b^2} = \underbrace{\frac{\rho_u}{\sigma_b^2} |\boldsymbol{\phi}_c^T \mathbf{z}_d|^2}_{\gamma_c} + 2\Re \left\{ \frac{\sqrt{\rho_u}}{\sigma_b^2} \boldsymbol{\phi}_c^T \mathbf{z}_d w_c \right\} + \frac{|w_c|^2}{\sigma_b^2}$
- **Select the IRS config** $C^* \in \mathcal{C}_{CB}$ such that $\hat{\gamma}_c \geq \gamma_0$, to get the desired diversity benefit.

Performance of CB-BSW for two schemes

● Fixed frame scheme: Select the IRS configuration. that gives the **best SNR**.

➔ Choice: $c^* = \arg \max_{c \in \mathcal{C}_c B} \{\hat{\gamma}_c \mid \hat{\gamma}_c \geq \gamma_0\}$

➔ If no configuration achieves the KPI, then we have an outage

● Flexible frame scheme: The BSW is stopped once the **KPI is met**.

➔ Choice of IRS config: $c^* = c \iff \hat{\gamma}_c \geq \gamma_0$

Algorithmic Errors in CB-BSW

- Outage occurs if **no configuration is selected** or a **wrong configuration is selected**.

- **Outage** probability: $p_{ae} = \mathcal{P}[\gamma_{c^*} \leq \gamma_0 | \hat{\gamma}_{c^*} > \gamma_0] + \mathcal{P}[\hat{\gamma}_c \leq \gamma_0, \forall c \in \mathcal{C}_{CB}]$
$$= \mathcal{P} \left[\gamma_{c^*} - \gamma_0 \leq \frac{|w_{c^*}|^2}{\sigma_b^2} + 2\Re \left\{ \frac{\sqrt{\rho_u}}{\sigma_b^2} \boldsymbol{\phi}_{c^*}^T \mathbf{z}_d w_{c^*} \right\} \right] + \mathcal{P}[\hat{\gamma}_1 \leq \gamma_0, \dots, \hat{\gamma}_{C_{CB}} \leq \gamma_0]$$

- **Chebyshev Bound** on the prob of 1st term:

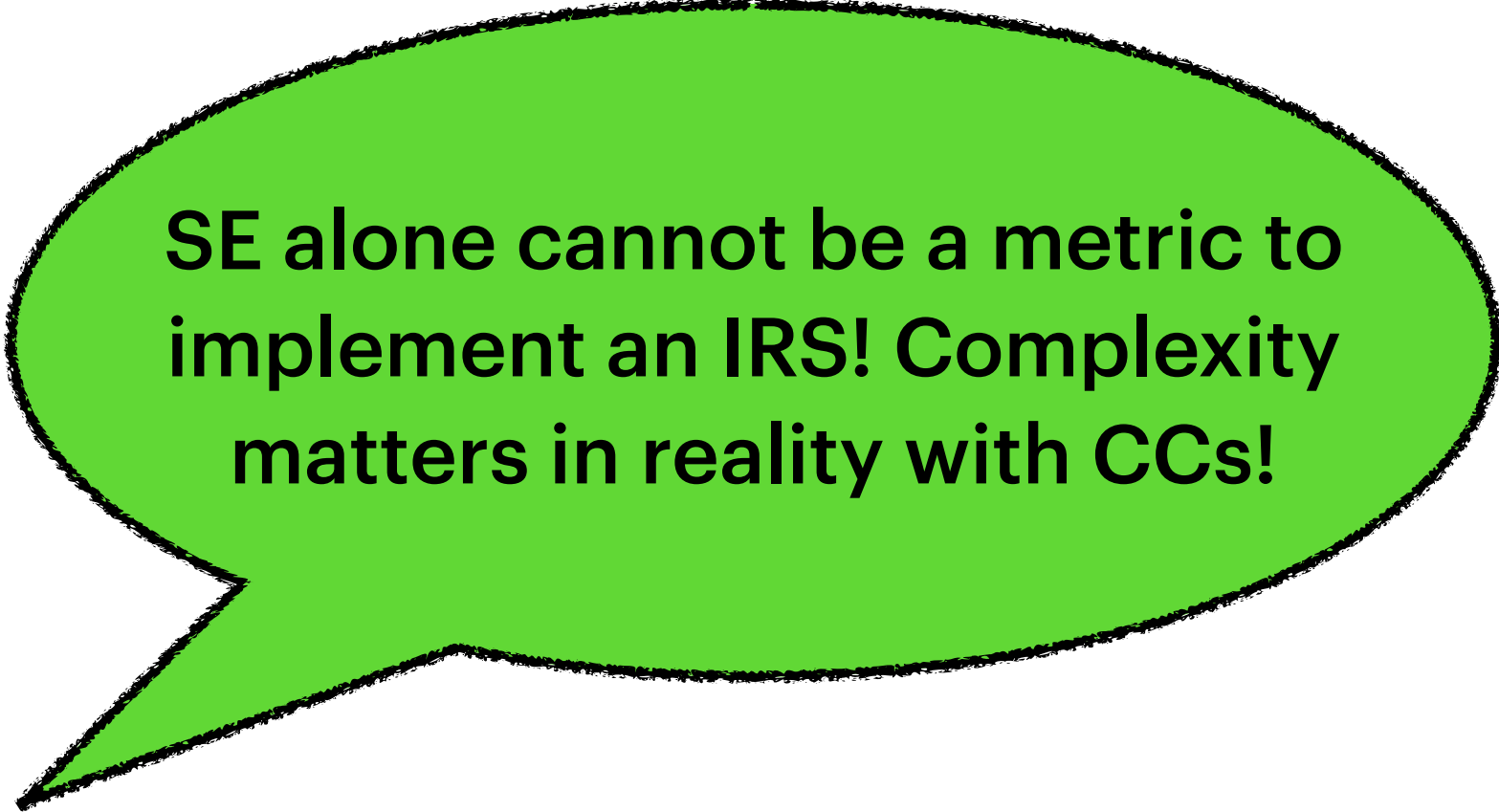
$$\mathcal{P} \left[\hat{\gamma}_{c^*} - \gamma_0 \leq \frac{|w_{c^*}|^2}{\sigma_b^2} + 2\Re \left\{ \frac{\sqrt{\rho_u}}{\sigma_b^2} \boldsymbol{\phi}_{c^*}^T \mathbf{z}_d w_{c^*} \right\} \right] \leq \frac{p^{-1}}{\hat{\gamma}_{c^*} - \gamma_0}$$

- Determining 2nd term depends on the CDF of estim. SNR, and changes with ch. model

- Fixed frame structure is **less susceptible to algorithmic errors** than flexible frames!

Trade-offs between OPT-CE and CB-BSW

- OPT-CE is a *fixed-rate* scheme - a global optimal rate
 - ➔ OPT-CE chooses the best config. and obtains the best rate
- CB-BSW is an *adaptive-rate* scheme
 - ➔ CB-BSW sets an SE a priori and finds an IRS config. - this SE may not be the max. value
- The SE performance of $\text{OPT-CE} \geq \text{SE of CB-BSW}$
- A *trade-off in complexity*
 - ◆ OPT-CE gives better perf. but incurs more complexity!
 - ➔ Multiple pilots scale with N
 - ➔ Phase optimization
 - ➔ Could transmit a new IRS vector to IRS-C also!



SE alone cannot be a metric to implement an IRS! Complexity matters in reality with CCs!

Impact of Control Channels

Performance Metrics

Measure comm. performances in the presence of control channels

● A **utility function** to measure

➔ **Overhead and the perf.** of comm. paradigms

➔ **Reliability** of CCs.

● **Goodput** R measures the performance of comm. paradigms along with algo. errors

➔ Evaluated as

$$R(\tau_{pay}, \eta) = \begin{cases} \frac{\tau_{pay}}{\tau} B_d \eta, & \text{with prob. } 1 - p_{ae}, \\ 0, & \text{with prob. } p_{ae}. \end{cases}$$

➔ $\eta = \eta_{CE}$ for OPT-CE and $\eta = \eta_{CB}$

➔ τ_{pay} is the total time for payload, a.k.a. the data transmission phase ($\tau_{pay} = \tau - \tau_{ini} - \tau_{alg} - \tau_{set}$)

➔ **Overhead time** depends on **pilot time** - τ_p , **optimization time** - τ_A , **control signaling from BS to UE** - $\tau_i^{(u)}$, **control signaling from BS to IRSC** - $\tau_i^{(r)}$, and **IRS switching time** - τ_s

Formulating the utility function

Modeling the reliability of CC

- Total number of control packets sent from BS: $P = P_u + P_r$
- If a packet is **decoded incorrectly, erroneous control** occurs.

- **Probability of correct control:**

$$p_{cc} = \prod_{k \in u, r} \prod_{i=1}^{P_k} (1 - p_i^{(k)})$$

- **Revised Goodput** expression:

$$R(\tau_{pay}, \eta) = \begin{cases} \frac{\tau_{pay}}{\tau} B_d \eta, & \text{with prob. } p_{cc} (1 - p_{ae}), \\ 0, & \text{otherwise.} \end{cases}$$

- **Utility function:**

$$U(\tau_{pay}, \eta) = p_{cc} (1 - p_{ae}) \left(1 - \frac{\tau_{ini} + \tau_{alg} + \tau_{set}}{\tau} \right) B_d \eta$$

Frame Timing diagrams

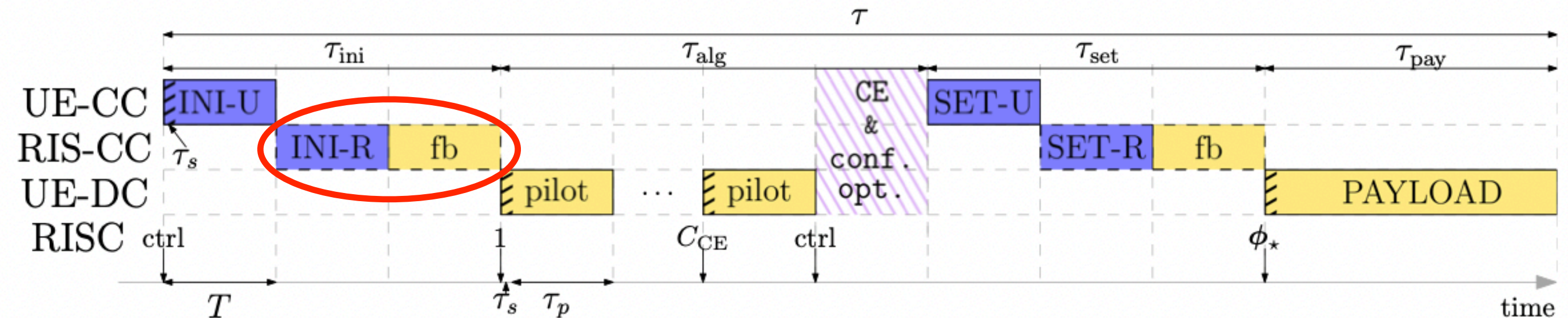
- Transmission time interval (TTI) = T , is the basic unit of time

- $P_r = 2, P_u = 2$.

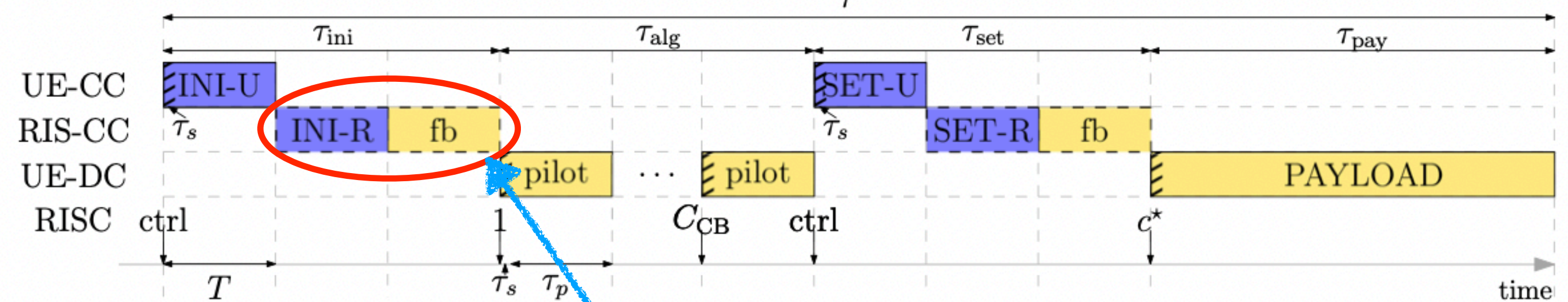
- Pilot length, $p = \left\lfloor \frac{T - \tau_s}{T_s} \right\rfloor$

- Algorithmic duration:

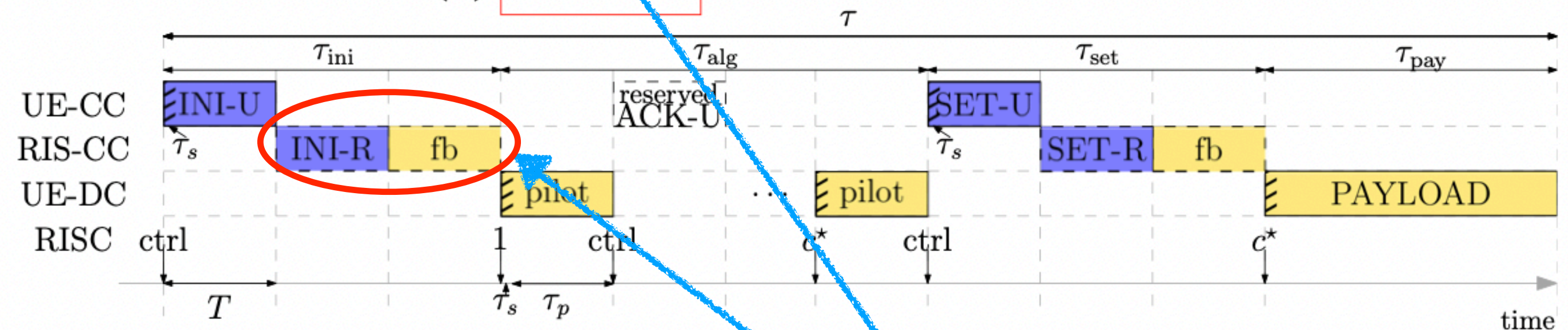
$$\tau_{alg} = \begin{cases} (C_{CE} + A)T, & \text{OPT-CE,} \\ C_{CB}T, & \text{CB-BSW fixed frame,} \\ (2c^* - 1)T, & \text{CB-BSW flexible frame.} \end{cases}$$



(a) OPT-CE



(b) CB-BSW: fixed frame structure



(c) CB-BSW: flexible frame structure

Not needed with
OB CC

Reliability Evaluation

⊙ Outage prob. of control pkt: $p_i^{(k)} = Pr \left\{ \log(1 + \Gamma_k) \leq \frac{b_i^{(k)}}{\tau_i^{(k)} B_k} \right\}, i = \{\text{INT}, \text{SET}\}, k \in \{u, r\}$

➡ More info bits, less time, and less BW lead to **high outage** probability.

⊙ Using the channel models, we can write $p_i^{(k)} = 1 - \exp \left[-\frac{1}{\lambda_k} \left(2^{b_i^{(k)}/(\tau_i^{(k)} B_k)} - 1 \right) \right]$

⊙ Correct control: $p_{cc} = \exp \left[\frac{1}{\lambda_u} \left(2 - \sum_{i=1}^2 2^{b_i^{(u)}/(\tau_i^{(u)} B_u)} \right) \right] \exp \left[\frac{1}{\lambda_r} \left(2 - \sum_{i=1}^2 2^{b_i^{(r)}/(\tau_i^{(r)} B_r)} \right) \right]$

➡ Note that $b_i^{(k)}$ and $\tau_i^{(k)}$ depend on the control info. requirement of a paradigm

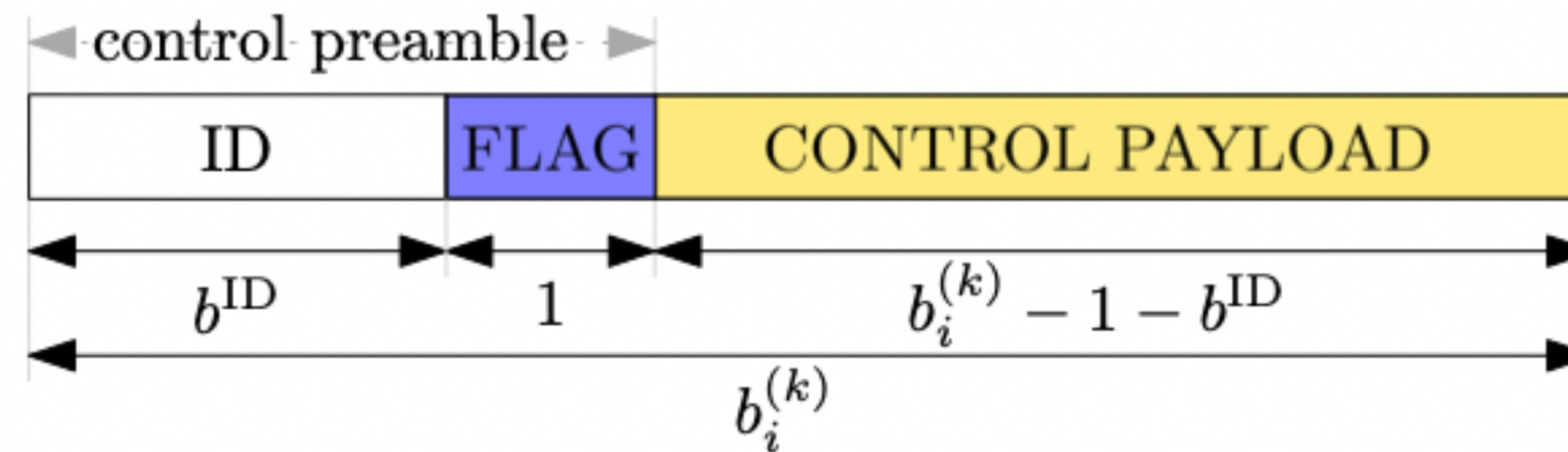
➡ So, different paradigms may require different SNRs λ_u & λ_r to maintain same p_{cc}

Actual reserved time for control packets $\tau_i^{(k)}$

- Recall that each control packet spans a TTI
- Due to the **switching time of IRS** elements, the actual **time for control pkt reduces**.
- For **IRS-CC, no such problems!**
 - ➡ For INT, SET: $\tau_1^{(r)} = \tau_2^{(r)} = T$
- For UE-CC, the actual usable time for control pkts are
 - ➡ For INT: $\tau_1^{(u)} = T - \tau_s$ for OPT-CE and CB-BSW
 - ➡ For SET: $\tau_2^{(u)} = T$ for OPT-CE (as long as $A \geq 1$)
 - ➡ For SET: $\tau_2^{(u)} = T - \tau_s$ for CB-BSW

Control Packet Contents

● General control packet structure:



➡ **Preamble** - destination ID

➡ **FLAG** - to say whether this is during INT or SET

➡ **Control payload** data depends on which comm. paradigm the system works in!

Control Payload for OPT-CE

- INI-U payload should contain frame time τ , Cardinality of C_{CE} , and guard time τ_s
- INI-R payload contains frame time τ , and set of config. $\mathcal{C}_{CE}$ to switch through
- SET-U payload contains the MCS index
- SET-R payload contains optimal IRS config. ϕ_* for data payload phase
- Summary of the control packet contents:

$$b_i^{(k)} = b^{ID} + 1 + \begin{cases} \underbrace{\lceil \tau/T \rceil}_{b^{frame}} + \underbrace{\lceil \tau_s/T_n \rceil}_{b^{guard}} + \underbrace{\lfloor \log_2(C) \rfloor}_{b^{conf}}, & k = u, i = 1, \\ b^{frame} + C_{CE} b^{conf}, & k = r, i = 1, \\ b^{SE} \text{ as per 5G std.} & k = u, i = 2, \\ Nb^{quant} & k = r, i = 2. \end{cases}$$

Control Payload for CB-BSW

- INT payload is are same as that in OPT-CE
- SET-U now does not contain any info
- SET-R contains the info of codeword c^* which gave the best SNR

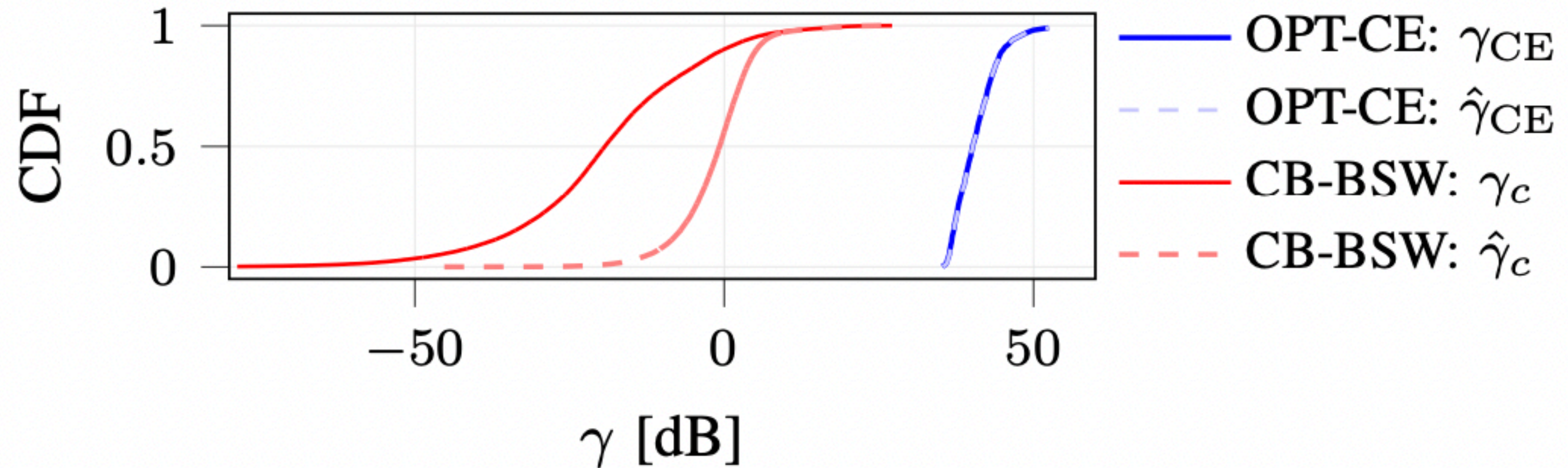
● Summary:

$$b_i^{(k)} = b^{ID} + 1 + \begin{cases} \underbrace{\lceil \tau/T \rceil}_{b^{frame}} + \underbrace{\lceil \tau_s/T_n \rceil}_{b^{guard}} + \underbrace{\lfloor \log_2(C) \rfloor}_{b^{conf}}, & k = u, i = 1, \\ b^{frame} + C_{CB} b^{conf}, & k = r, i = 1, \\ 0, & k = u, i = 2, \\ b^{conf} & k = r, i = 2. \end{cases}$$

- CB-BSW needs less control payload than OPT-CE, so reliability of CC is more for CB-BSW!

Numerical Studies

Performance in error-free CCs



- OPT-CE is **more robust** to errors
- CB-BSW **suffers high outage** events (due to algorithmic errors)
- **Seems there is a typo in the legend!**

Optimal target SNR exists in CB-BSW

- Smaller γ_θ (threshold in CB-BSW):
 - ➔ C_{CB} is small in Fixed frame
 - ➔ **Wrong codeword may be selected** in both variants
- Larger γ_θ
 - ➔ **No codeword is selected** - outage is more!
- So, **optimal γ_0 exists** for any τ
- Goodput of **OB-CC is slightly better than IB-CC**
- 'Seems' no variation across τ
 - ➔ But, they **should increase due to longer payload**

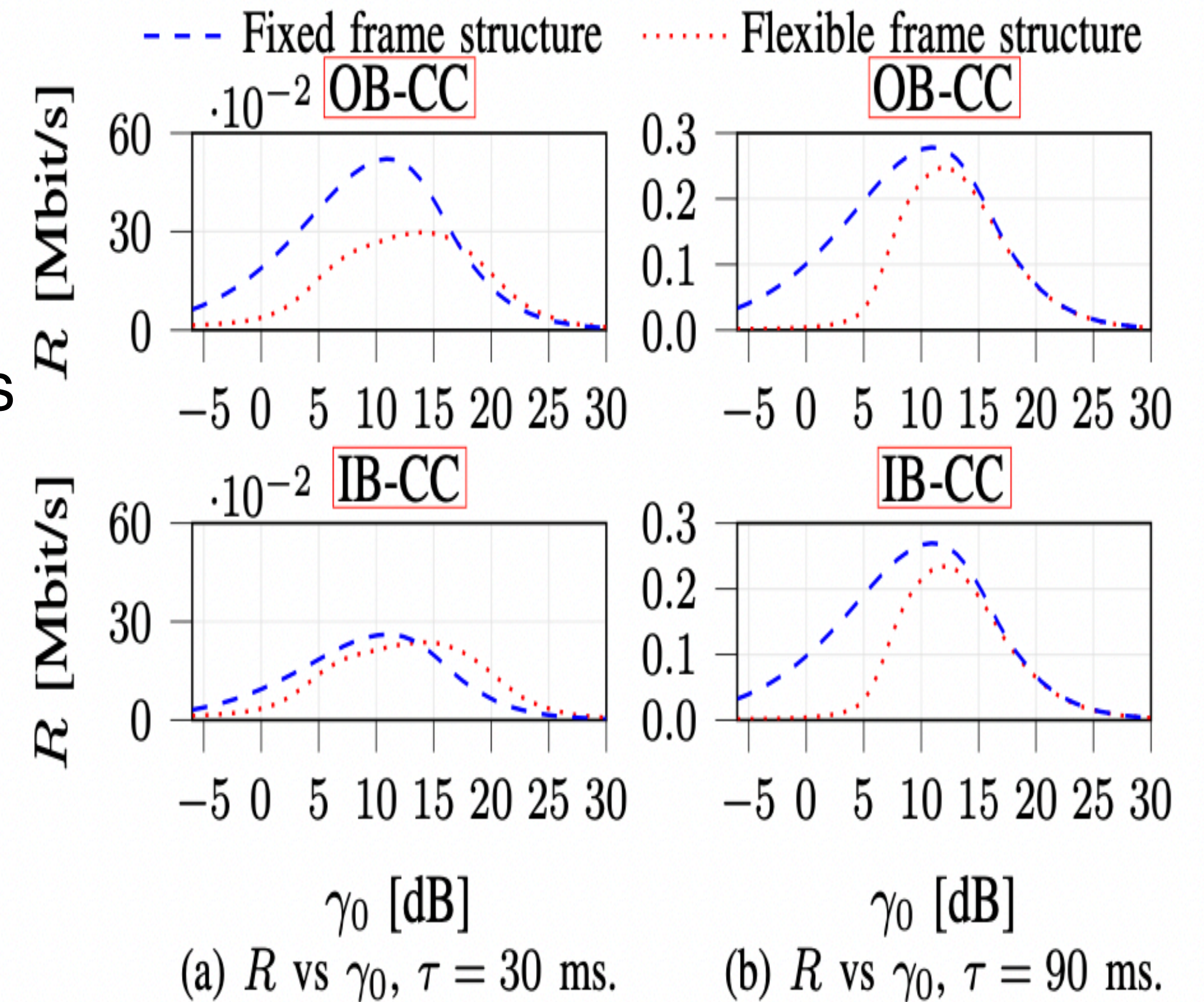


Fig. 6: Analysis of the target SNR for the CB-BSW paradigm (note the different scale in Fig. 6a).

When is OPT-CE good? When is CB-BSW good?

- When τ is small, R of OPT-CE performs poorly
 - ➔ Not enough pilots to obtain accurate CSI
- As τ increases, OPT-CE outperforms CB-BSW
- For larger τ , R of CB-BSW increases due to more payload
- The impact of CC on R is negligible
- However, the CDF of R depends on CC
 - ➔ More control in OPT-CE - so OB-CC is better
 - ➔ CB-BSW depends less on CC
 - ➔ CBS-BW is prone to algo. errors for 50% time

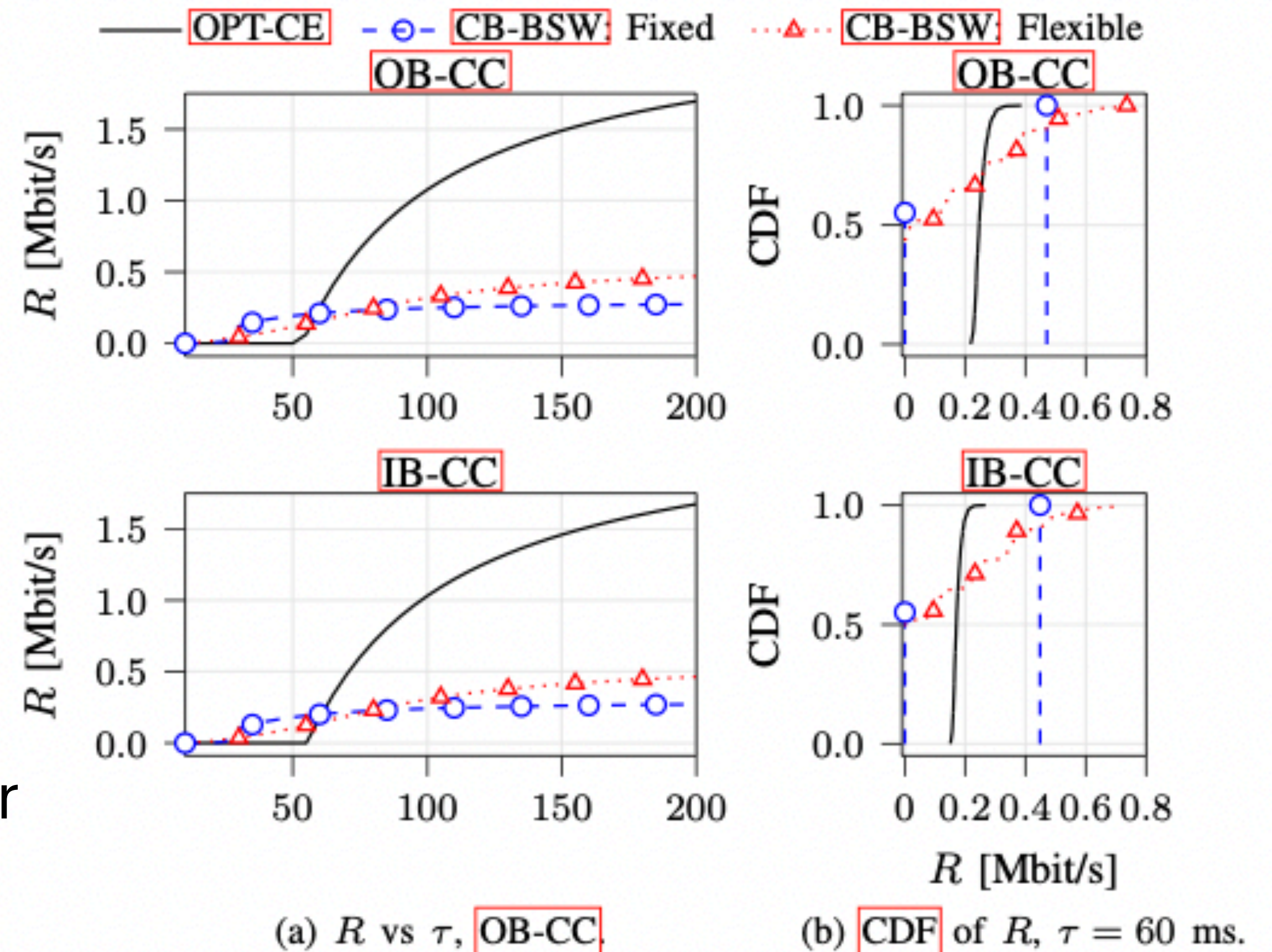


Fig. 7: Analysis of the goodput performance.

Utility function vs. outage in CCs

● Recall:

$$U(\tau_{pay}, \eta) = p_{cc}(1 - p_{ae}) \left(1 - \frac{\tau_{ini} + \tau_{alg} + \tau_{set}}{\tau} \right) B_d \eta$$

- OPT-CE gives the **best utility** when τ is sufficiently **large**
- The impact of **CC outage** is **uniform across IB and OB**
- Utility is a **weak function of loss in time** due to IB-CC.

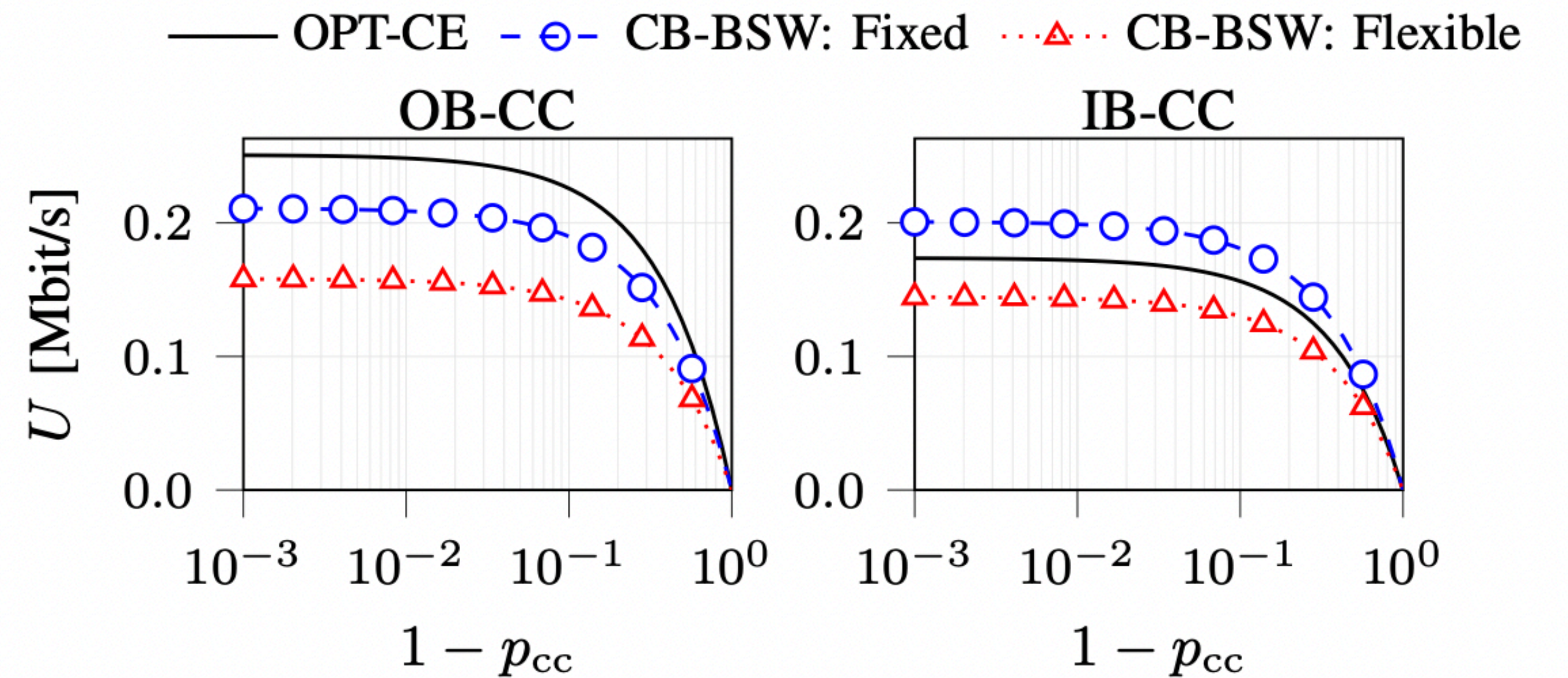


Fig. 8: Analysis of the utility function vs. erroneous control probability for $\tau = 60$ ms.

Impact of CC SNR on perf.

- The reliability of the CCs is the **same** for **OPT-CE & CB-BSW** in **OB-CC**
 - ➔ The control info is almost the same in UE-CC for both paradigms
- The required **CC SNRs are slightly higher** for OPT-CE than for CB-BSW
 - ➔ More control payload in the former

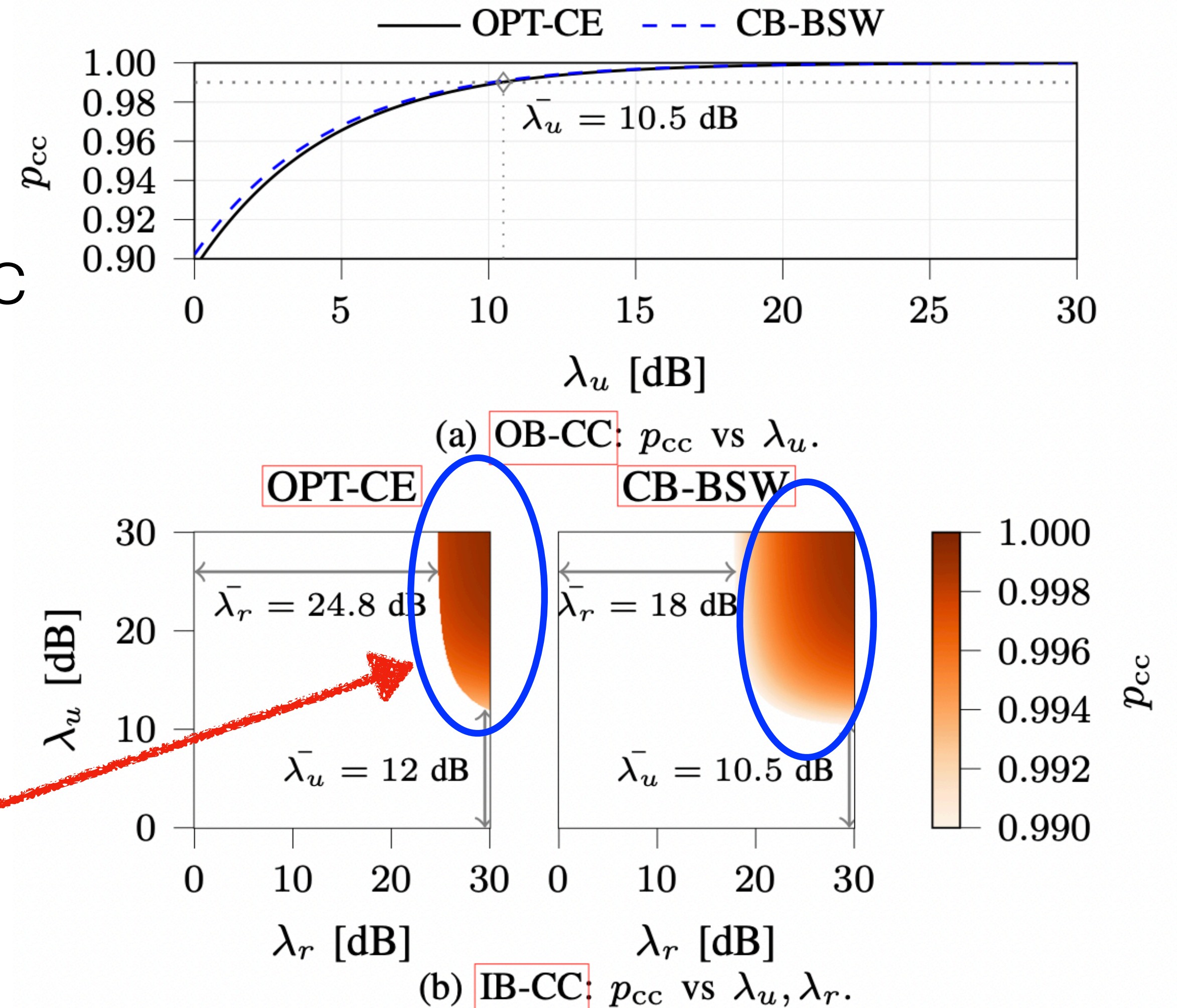


Fig. 9: Impact of the CC SNR on the reliability performance.

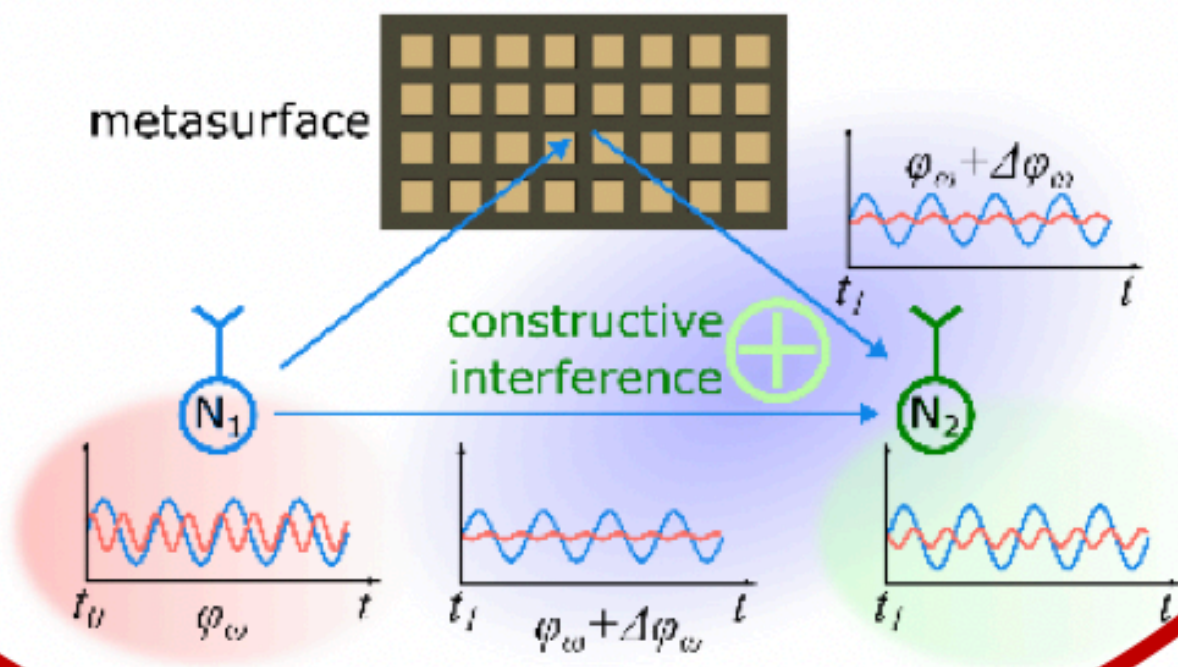
Summary

- Control channels have to be **designed robustly** for promised IRS benefits
- Generally, the CCs mainly **impact the reliability of txn more than the loss in payload**
 - ➔ The control **data duration directly reflects in the outage** of CC links
 - ➔ However, for OPT-CE and IB-CC, control payload also deteriorates the performance
 - ➔ **Outage** (algo+CC) causes **severe degradation than loss in payload time**
- **OPT-CE is best preferred** when
 - ➔ **CC SNRs are high** (e.g., IB-CC for IRS-CC)
 - ➔ **Coherence time is larger** for longer frames
- **CB-BSW is best preferred** when
 - ➔ **Coherence time is shorter** so that outages are prevented to the maximum extent
- Different paradigms may require **different strengths of CC for equal reliability**

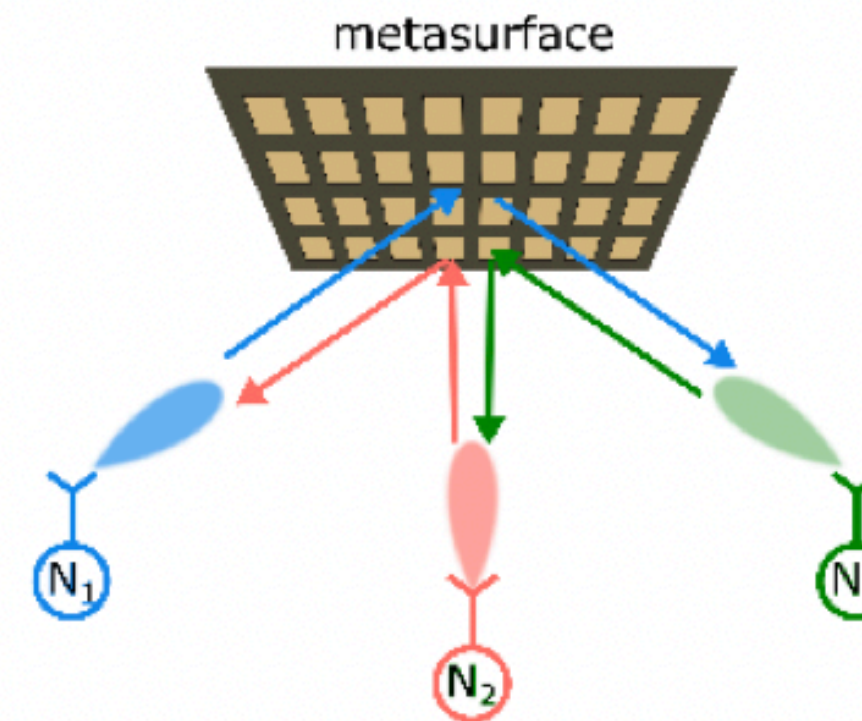
Meta-surfaces for Next-gen wireless!

RIS

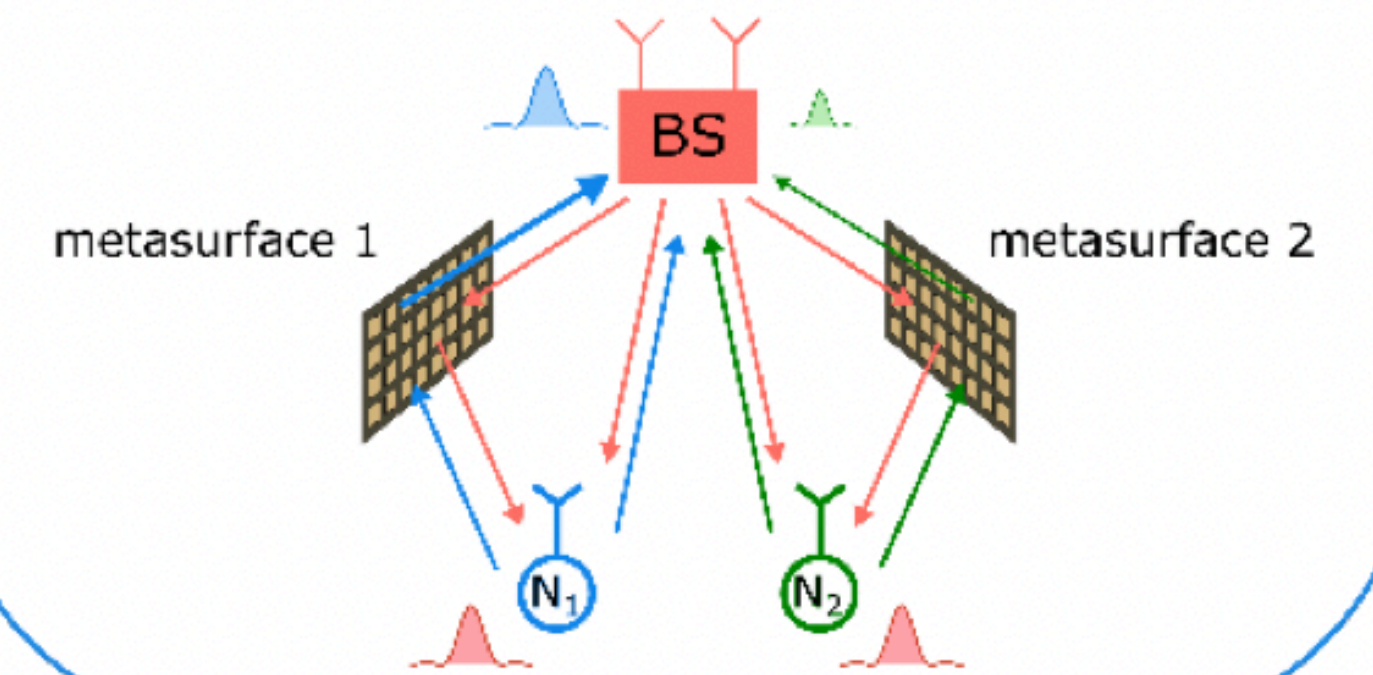
a Multi-path connection



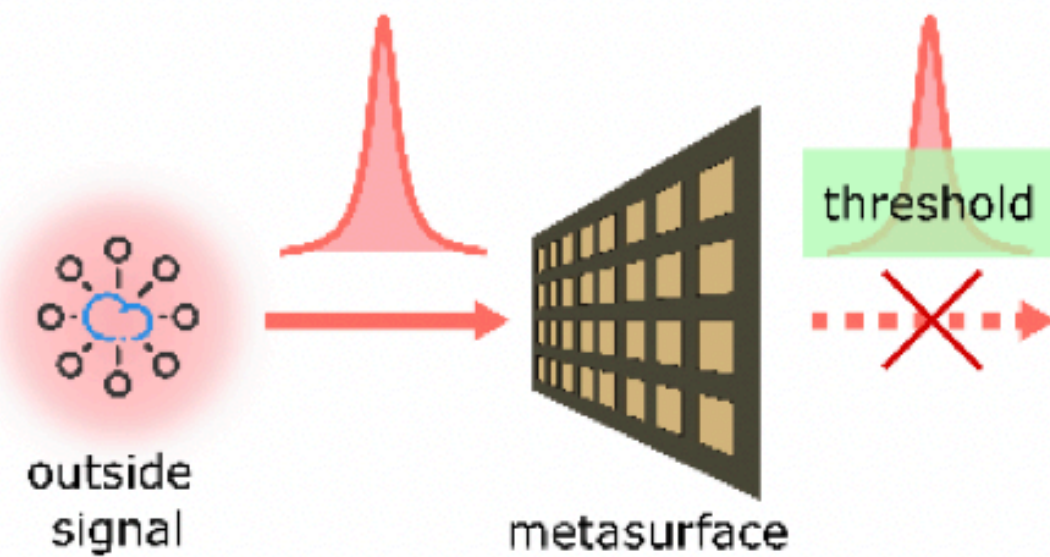
b In-band full duplex communication



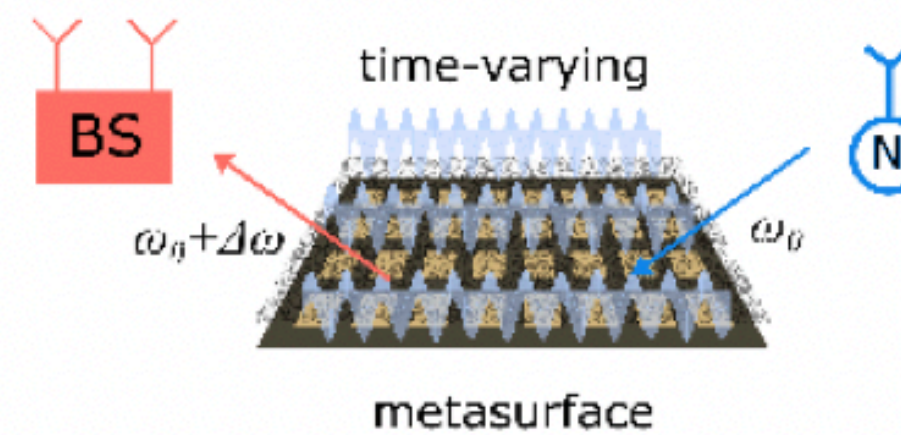
c Two-way multi-user communication



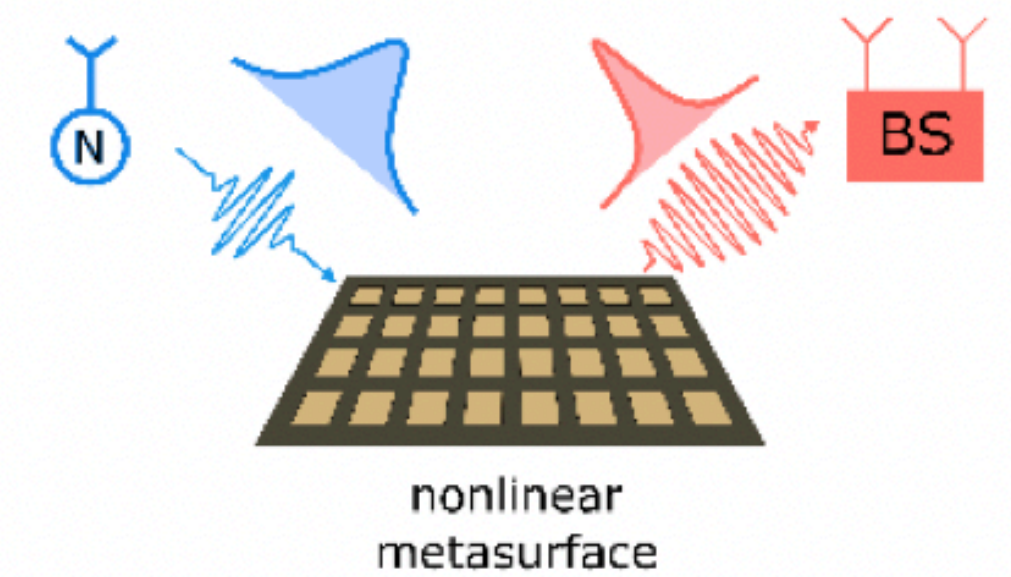
d Signal limiting



e Doppler cloak



f Bandwidth narrowing



© References:

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Thank You! :)